

Merkel cell tumour: clinical behaviour and treatment

We have reviewed 30 reports of Merkel cell tumour and described a further five cases in order to establish a database and from this more clearly define the biology of this tumour, prognostic factors that govern outcome, and optimal management. After excision alone of the primary lesion, local recurrence occurred in 39 per cent of patients and regional failure occurred in 46 per cent. In contrast, in patients treated by excision plus prophylactic treatment (adjuvant node dissection and/or adjuvant radiation), local recurrence occurred in 26 per cent and regional failure in 22 per cent. Locoregional recurrence carried an ominous significance with 67 per cent of patients subsequently dying of the disease. For patients who either presented with regional disease or later developed regional disease, the best outcome (44 per cent survival with mean follow-up of 40 months) was obtained following treatment by therapeutic node dissection with or without radiation. In contrast, treatment of regional disease with radiation alone was associated with only a 20 per cent survival rate. Unfavourable prognostic factors included young age, lesions sited in the head and neck or trunk, male sex, and the presence of locoregional failure and/or systemic disease. We conclude that Merkel cell tumours behave in a similar manner to the aggressive variants of melanoma and that minimal treatment consists of wide surgical resection of the primary lesion (with a margin of 2.5–3 cm) coupled with resection and probably also radiation of regional disease if present. In addition, consideration should be given to prophylactic node dissection in node negative patients, especially in those patients with unfavourable prognostic factors.

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Merkel cell tumour is an unusual neuroendocrine tumour. Typically skin lesions present on the limbs or head and neck, are pink or grey and, because they arise in the dermis rather than the epidermis, rarely ulcerate^{1–10}. The histology of Merkel cell tumour is now well described^{1–5}, and more recently the ultrastructural details have been resolved^{10–13}. However, the clinical behaviour of this tumour is not well described, and there is no agreement regarding optimal treatment as the majority of clinical reports concern small numbers of patients; in many of these reports the emphasis has been on pathology and diagnosis rather than treatment.

We have reviewed the details of 234 patients with Merkel cell tumour: five cases of our own and 229 cases reported in 30 papers^{1–30}. We have focused our review on clinical presentation and tumour behaviour, and have attempted to define prognostic factors as well as a strategy for treatment.

Patients and methods

We have reviewed the data presented in 30 reported series of patients with Merkel cell tumour^{1–30}. This accounted for 293 cases in total but in only 229 cases was there sufficient information available to incorporate into a database. We have also reported five new cases, bringing the total to 234 cases (Table 1).

Data were collected with particular reference to age, sex and race of patient, mode of presentation, diagnosis, primary management, salvage therapy and outcome. Prognostic factors influencing outcome have been defined and the statistical significance of each factor has been assessed using a χ^2 analysis.

The five patients seen in Auckland had a mean age of 75 years (range 57–87 years) and all but one were women. In two cases the tumour was in the head and neck, and in three cases in the limbs. Three patients had nodal disease at presentation, and one patient had systemic disease. One patient previously had skin cancer and one had dural ectasia. Two patients underwent wide local excision alone and the three who presented with regional disease were treated by wide local excision, therapeutic lymph node dissection and adjuvant radiation. Four of the five patients are alive and well with a mean follow-up of 6 months.

Results

The primary site for tumours in the 234 patients was head and neck (108 patients), leg (77 patients), arm (42 patients) and trunk (seven patients). There were 129 women with a mean age of 70 years (range 23–92 years) and 105 men with a mean age of 68 years (range 23–92 years; see Table 1).

Table 1 Details of patients with Merkel cell tumours

Author (year)	Reference	n	Mean age	M:F	Head and neck	Trunk	Limbs
Toker (1972)	1	5	70	2:3	2	1	2
Van Dyk and Ten Seldam (1975)	2	1	72	0:1	1	—	—
Tang and Toker (1978)	3	9	61	4:5	2	1	6
Taxy <i>et al.</i> (1980)	4	1	74	0:1	1	—	—
De Wolf Peeters <i>et al.</i> (1980)	5	5	69	0:5	3	—	2
Sibley <i>et al.</i> (1980)	6	2	80	1:1	2	—	—
Johannessen and Gould (1980)	7	1	78	0:1	1	—	—
Ecker <i>et al.</i> (1982)	8	6	56	1:5	3	—	3
Pilotti <i>et al.</i> (1982)	9	10	66	2:8	3	1	6
Warner <i>et al.</i> (1983)	10	6	62	3:3	1	2	3
Beyer <i>et al.</i> (1933)	11	1	80	0:1	1	—	—
Rustin <i>et al.</i> (1983)	12	1	75	0:1	—	—	1
Frigerio <i>et al.</i> (1983)	13	15	70	3:12	6	1	8
Goepfert <i>et al.</i> (1974)	14	29	66	23:6	29	—	—
Pettinato <i>et al.</i> (1984)	15	1	61	0:1	—	—	1
Sibley <i>et al.</i> (1985)	16	43	70	21:22	17	2	24
Schnitt <i>et al.</i> (1966)	17	1	60	1:0	—	—	1
O'Rourke and Bell (1986)	18	1	90	0:1	1	—	—
Meland and Jackson (1986)	19	23	63	15:8	10	4	9
Raaf <i>et al.</i> (1986)	20	4	68	2:2	—	1	3
Leong and Dixon (1986)	21	13	70	8:5	7	1	5
Cotlar <i>et al.</i> (1986)	22	8	67	7:1	5	—	3
O'Rourke and Bourne (1987)	23	3	62	3:0	3	—	—
O'Brien <i>et al.</i> (1987)	24	11	72	7:4	8	—	3
Bourne and O'Rourke (1988)	25	13	70	9:4	11	—	2
Tennvall <i>et al.</i> (1989)	26	18	70	10:8	9	1	8
Pacella <i>et al.</i> (1989)	27	20	65	8:12	15	1	4
Shaw and Rumball (1990)		5	75	1:4	2	—	3
Wick <i>et al.</i> 1983	28	13	62	8:5	7	—	6
Kirkham and Isaacson (1983)	29	3	79	1:2	2	—	1
Kroll and Toker (1982)	30	21	69	11:10	9	—	12
Total		293	70	151:141	161	16	116

Stage of disease

Twenty-three patients had regional disease on presentation. Of the 108 patients with head and neck tumours, eight (7 per cent) had nodal disease at presentation, as did five of 42 (12 per cent) patients with arm lesions, and 10 of the 77 (13 per cent) patients with leg lesions. Three patients had systemic disease at presentation, and the remaining 208 had localized disease.

Node negative patients

Initial treatment. There were 208 patients in this group. Wide local excision alone was performed in 181, and in 27 patients surgical treatment of the primary lesion was coupled with prophylactic nodal dissection, nodal radiation, or a combination of the two.

After wide local excision alone, 83 of the 181 patients developed regional failure (46 per cent), and 71 (39 per cent) also developed local recurrence. Seventy patients died of the disease (39 per cent), and the 111 surviving patients were followed for an average of 25 months.

In contrast, in the patients who underwent prophylactic node dissection and/or radiation, locoregional recurrence was less common. Of the 27 patients, seven (26 per cent) developed local recurrence and six (22 per cent) also developed regional failure. Eight patients (30 per cent) died of the disease and the remaining 19 patients (70 per cent) were alive after a mean follow-up of 40 months (see Table 2).

Table 2 Treatment and outcome in node-negative patients

	n	Alive	Local failure	Regional failure	Death
Excision alone	181	110 (60)	71 (39)	83 (46)	70 (39)
Excision + adjuvant treatment	27	19 (70)	7 (26)	6 (22)	8 (30)

Values in parentheses are percentages. Adjuvant treatment: prophylactic node dissection or nodal radiation in addition to excision of primary lesion. Mean follow-up for survivors: 28 months. Mean time from initial treatment until recurrence: 7 months for local recurrence, 12 months for regional recurrence. Mean time between locoregional failure and death: 11 months

Subsequent treatment of delayed regional disease. Three patients did not undergo any treatment, and all died (mean survival 3 months). In addition, six patients were treated with a variety of chemotherapeutic agents either alone or in combination, and all died (mean survival 13 months).

Radiation alone was used in 25 patients. Five (20 per cent) were alive after a mean follow-up of 40 months, and 80 per cent (20 of 25) died of the disease (mean survival 14 months) after developing regional failure.

Fifty-five patients were treated by lymphadenectomy either alone or in combination with adjuvant radiation. Thirty-two

Table 3 Treatment versus outcome in node-positive patients

	n	Alive	Dead
No treatment	4	0	4 (100)
Chemotherapy	7	0	7 (100)
Radiation	30	6 (20)	24 (80)
Node dissection with or without radiation	71	31 (44)	40 (56)

Values in parentheses are percentages. Note: both patients who presented with nodal disease and patients who subsequently developed nodal disease are included

of 55 (58 per cent) died of the disease (mean survival 13 months) after developing regional failure, and 23 of 55 (42 per cent) were alive after a mean follow-up of 54 months.

Significance of local and regional failure. Both local recurrence and regional failure were associated with a high chance of death occurring. Seventy-eight of the 208 patients who were initially node negative died of the disease (38 per cent). Fifty-seven (73 per cent) had previously developed local recurrence, 61 (78 per cent) had previously developed regional failure, and all had metastatic disease.

Patients who presented with synchronous or metachronous regional disease

Treatment. Local excision with radiation was used in five of the 23 patients who had regional disease on presentation. Excision of the primary lesion plus lymph node dissection was employed in 11, excision of the primary lesion with node dissection and radiation was performed in five patients, one patient was not treated, and one patient underwent excision of the primary lesion plus chemotherapy.

Overall, 14 patients (61 per cent) died of the disease, with a mean survival time of 17 months. The remaining nine patients were alive after a mean follow-up of 12 months.

Four of the five patients treated by excision plus radiation died of the disease, while seven of 11 (64 per cent) patients treated by excision and node dissection died of the disease. Of the five patients treated by excision, node dissection and radiation, one died at 8 months, and the remaining four were alive after a mean follow-up of 18 months.

Results. The overall influence of individual treatment regimens for patients who either presented with regional disease or subsequently developed regional failure is summarized in Table 3.

Treatment of systemic disease

Three patients presented with systemic disease and 81 patients developed systemic disease after initial treatment. The outcome after treatment of systemic disease was poor. Twenty-six patients were not treated and the mean time from development of metastases until death was 4 months. Twenty-seven patients were treated by radiation, 14 with a variety of chemotherapeutic agents, and 17 were treated by both radiation and chemotherapy. The mean times from development of metastases until death were 8 months, 8 months and 6 months respectively.

Prognostic factors

Sex was a prognostic factor in that 47 per cent (49 of 105) of men died *versus* 34 per cent (44 of 129) of women.

Anatomical site also influenced outcome. Twenty-nine per cent (35 of 119) of patients with limb lesions died and 45 per cent (52 of 115) of patients with trunk or head and neck lesions died of the disease.

Age was a prognostic factor in that 40 per cent (67 of 166) of patients aged >60 years, 53 per cent (25 of 47) of patients aged 40–60 and 60 per cent (three of five) of patients aged <40 years died of the disease.

Nodal status was also predictive. Seventy-five of the 112 (67 per cent) patients who either presented with nodal disease or subsequently developed regional failure died. In contrast, of the patients who did not develop regional disease, only 27 of 115 (23 per cent) died.

Systemic disease was prognostic as all but one of the 84 patients with systemic disease died of the disease; the remaining patient had a spontaneous remission.

Local recurrence was prognostic given that 57 of 71 patients (80 per cent) who developed local recurrence died of the disease (see Table 4).

Discussion

One striking feature of Merkel cell tumour is the similarity of its biology to that of malignant melanoma. This is no surprise as both cancers occur most commonly in the skin, and both are of neural crest origin^{1–5}. As treatment regimens for Merkel cell tumour are still evolving, we have attempted to fill some of the gaps in our knowledge by drawing on what is known for melanoma.

The five additional patients we have reported demonstrate some new aspects. One of our patients with a large head and neck tumour was Polynesian and we are unaware of any other cases reported in non-Caucasians. This is consistent with the racial distribution of melanoma. In our review of melanoma in Auckland (the largest Polynesian city) only 1.5 per cent of melanomas occurred in non-Caucasians³¹. Another of our patients had the associate neural crest defect of dural ectasia. This finding, along with the observations of others that: (a) Merkel cell tumour may develop in association with other types of skin cancer of neural crest origin^{13,14,22,23,27}, (b) that Merkel cell tumours have been associated with calcitonin secretion (calcitonin is normally secreted by the neural-crest derived parathyroid glands), and (c) that there may be an association of some endocrine tumours on the basis of common neural crest origin development (multiple endocrine neoplasia syndromes types 1–3) raises the possibility that there may be an association between Merkel cell tumours and other tumours of neural crest origin.

Our review demonstrates that the most common site for the tumour is the head and neck, followed by the limbs, and that the average age of patients at presentation is approximately 70 years, with females predominating. These demographic features differ from those of melanoma in the sense that melanoma, although more common in females, is most commonly seen in the limbs (only 15 per cent of cases occur in the head and neck)³², and the average age at presentation is younger³¹.

Table 4 Prognostic factors in patients with Merkel cell tumour

	n	Deaths	χ^2	P
Sex				
Male	105	49		
Female	129	44	3.8	<0.05
Site				
Head and neck, trunk	115	52		
Limbs	119	35	7.1	<0.01
Age				
≤60 years	67	37		
>60 years	166	67	5.3	<0.02
Nodes				
Not involved	115	27		
At presentation	23	14	14	<0.001
Delayed	89	61	>20	<0.001
Local failure				
Present	71	57		
Not present	121	35	>20	<0.001
Systemic disease				
Present	84	83		
Not present	150	5	>20	<0.001

The prognostic features of Merkel cell tumour (including the age and sex of the patient, site of the primary lesion, and the presence of local or regional recurrence or systemic disease) can also be compared with equivalent data from melanoma patients. In Merkel cell tumour the outcome for younger patients was less favourable than for older patients, while for melanoma the opposite has been recorded³³. For Merkel cell tumours the prognosis was less favourable for lesions in the head and neck or trunk than for limb lesions, a finding that parallels the situation in the melanoma^{31,34}. As is the case with melanoma, the presence of regional or systemic disease carried an ominous prognosis³², and there is no effective treatment for systemic disease in either condition. The 5-year survival rate for patients with melanoma with regional disease is approximately 30 per cent³⁵, and the situation for patients with Merkel cell tumour appears to be similar in that 33 per cent of patients with regional disease were alive after a mean follow-up of 51 months. The outlook for patients with Merkel cell tumour and patients with melanoma with systemic disease also appears to be similar. At present there is no effective chemotherapy regimen for melanoma³⁶. Response rates of up to 25 per cent are reported but the majority of patients with systemic melanoma are dead within 6–9 months³⁶. The pattern of survival for the patients with Merkel cell tumour who have systemic disease was similar irrespective of whether they were treated with chemotherapy and/or radiation, or managed expectantly.

Overall the outcome for patients with Merkel cell tumour is poor with approximately 35 per cent of patients dying of disease, and approximately 30 per cent of the patients reported as alive having been followed for less than 1 year and therefore still at risk. The two major problems associated with this high death rate are the development of local or regional recurrence. Half of the patients developed either local or regional failure or a combination of the two, and this was associated with death in more than 60 per cent of cases. Again this is consistent with the situation which pertains with melanoma^{31,32}.

The results of treatment of patients with regional disease were dependent on the treatment regimen employed, and the poor overall results underline the importance of avoiding this problem if possible. Management with radiation alone was unsatisfactory as 80 per cent of patients died at a mean of 14 months after developing nodal disease. In contrast, when patients with regional disease were treated by lymph node dissection alone or in combination with radiation, 44 per cent were alive after a mean follow-up of 40 months. These results parallel those following the treatment of many types of head and neck cancer, or in regional head and neck melanoma where results after the use of radiation alone to control regional disease are suboptimal^{36,37}. We have previously reported that in patients with regional disease from squamous cell carcinoma of tongue³⁸ or lip³⁹, the results after radiation alone are poor. In addition, it is generally accepted that management of regional melanoma with lymph node dissection³⁵ results in approximately 30 per cent of patients surviving a further 5 years; in contrast, radiation of involved nodes although often providing symptomatic relief seldom results in 5-year survival⁴⁰.

Treatment of patients with Merkel cell tumour who have systemic disease is also difficult. Patients receiving no treatment survived for an average of 3–4 months after developing systemic disease, and those receiving radiation, chemotherapy, or radiation plus chemotherapy survived for less than 8 months. Again these results parallel those for melanoma where the overall response rate for chemotherapy is low and long-term survivors are few⁴¹; although radiation is effective in palliating symptoms resulting from systemic disease it has little effect in terms of prolongation of life⁴⁰.

Because local and regional failure were two major factors predisposing to death, and because the results of salvage treatment were poor, an optimal treatment regimen should include measures to minimize locoregional failure. In this respect guidance can be obtained from our understanding of

melanoma, for which clearer treatment details have been established⁴².

The 35 per cent incidence of local recurrence of Merkel cell tumour after excision alone was almost seven times higher than that noted in our review of almost 1000 melanoma patients (4.4 per cent³¹). Even if the patients with level 5 melanomas were considered as a subgroup, the incidence of local recurrence was still less than half that seen in the patients with Merkel cell tumour. On this basis it is reasonable to assess the results of the initial treatment of Merkel cell tumour in the light of what would be expected for level 5 melanomas or for melanomas >2–3 mm in thickness.

Few records of Merkel cell tumour supply detail of resection margins and this may explain the high incidence of local failure. For melanoma there appears to be little advantage in terms of minimizing local recurrence by taking a margin of >3 cm even for melanomas that are thick and known to have an unfavourable outlook^{43,44}. In our previous review, the incidence of local recurrence for head and neck melanoma was 50 per cent higher than that for limb lesions^{31,32}, and this was largely on the basis of narrow margins being used in the head and neck for cosmetic reasons. For patients with Merkel cell tumour, nearly half of the lesions occur in the head and neck region, and the incidence of local recurrence for head and neck patients (44 per cent) was almost twice the value seen in patients with limb lesions (23 per cent). It is therefore likely that in many instances the resection margin in the head and neck patients may have been short of the desired 2.5–3.0 cm of clearance that would be optimal for a thick melanoma. Thus, rather than regarding the head and neck as an unfavourable anatomical site for Merkel cell tumours *per se*, the less favourable outlook may reflect inadequate local treatment.

The situation regarding prophylactic lymph node dissection in melanoma is debated. Although two prospective studies^{45,46} have shown no advantage for patients with limb lesions, both have been criticized as having interpretive problems, and there are retrospective data from thousands of patients indicating a survival advantage in patients with lesions 2–4 mm of thickness⁴⁷. Our review indicates that prophylactic treatment of regional nodes (either with surgery or with surgery coupled with radiation) is advantageous in patients with Merkel cell tumours. The incidence of loco-regional failure was almost halved in the patients who received nodal treatment in addition to excision alone.

Thus, when considering a treatment strategy for Merkel cell tumour, the following factors should be borne in mind:

1. This tumour behaves as if it were an aggressive variant of melanoma.
2. A local recurrence rate of 33 per cent indicates that an adequate margin of clearance (probably at least 2.5–3 cm), that is similar to that required for a thick melanoma, must be obtained.
3. Regional failure occurs in almost 50 per cent of patients with Merkel cell tumour after excision of the primary lesion alone, indicating a need to consider prophylactic treatment of regional nodes. Based on current knowledge of melanoma, prophylactic treatment should be considered in patients who have lesions likely to be associated with subsequent regional failure. This includes young patients, men, and patients with primary lesions in the head and neck.
4. Our review indicates that radiation alone is not optimal therapy for Merkel cell tumours, although some case reports describe complete resolution after radical radiotherapy⁴². However, adjuvant radiation following surgical resection appears to be effective in minimizing loco-regional recurrence.
5. As for melanoma, there is no effective chemotherapy for Merkel cell tumour, and management of systemic disease is probably best accomplished with palliative radiotherapy.

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